# EXPERIMENT 1: SIMULATION OF DFA

## Aim

To write a C program to simulate a Deterministic Finite Automaton (DFA) that accepts strings starting with a and ending with a.

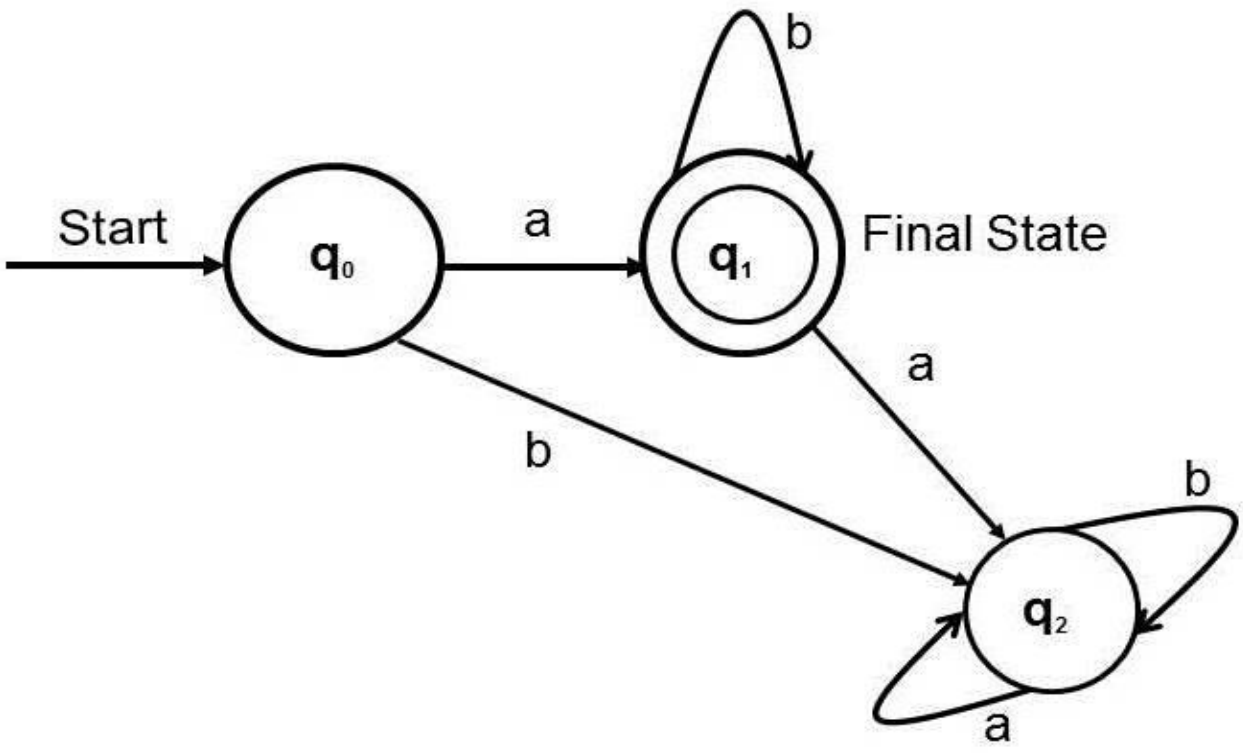
## Procedure / Algorithm

1. Start the program.
2. Read a string from the user.
3. Find the length of the string.
4. Check whether the first and last characters are a.
5. If both conditions are satisfied, print "String Accepted"; otherwise, print "String Rejected".
6. Stop the program.

## Explanation:

A Deterministic Finite Automaton (DFA) accepts a string if it starts with a and ends with a. The input alphabet is Sigma={a,b}. Each state has exactly one transition for each input symbol.

## Sample Design:



**Transition Table:**

| Present state | Input a | Input b |
|---|---|---|
| → q_0 | q_1 | q_d |
| q_1 | q_1 | q_2 |
| q_2 | q_1 | q_2 |
| q_d | q_d | q_d |

## Code:

```c
#include <stdio.h>
#include <string.h>
int main()
{
    char s[100];
    int n;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (s[0] == 'a' && s[n - 1] == 'a')
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

```
Enter a string: aabaa
String Accepted
```

## Input / Output:

Input: Text Enter a string: aba Output:

Text :String Accepted

## Result:

Thus, the C program to simulate a DFA that accepts strings starting with a and ending with a was executed successfully.